

JEE Main - 19 | JEE - 2022

Date: 06/02/2022

Maximum Marks: 360

Timing: 09:00 AM to 09:00 PM

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **360**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **10 Numerical Value Type Questions Out of which ONLY 5 questions have to be attempted**.
The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

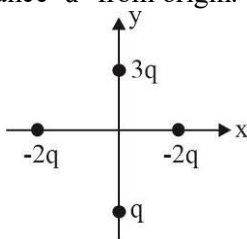
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

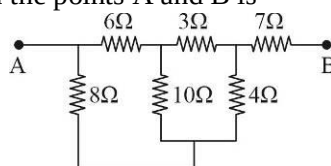
PART - I : PHYSICS**120 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

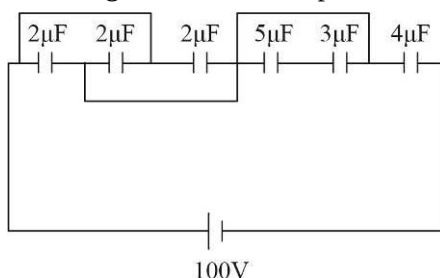
- An ideal choke takes a current of 8 A when connected to an a.c. source of 100 volt and 50Hz. A pure resistor under the same conditions takes a current of 10A. If two are connected in series to an a.c. supply of 100V and 40Hz, then the current in the series combination of above resistor and inductor is
 (A) 10A (B) 8A (C) $5\sqrt{2}$ A (D) $10\sqrt{2}$ A
- In a Young's double slit experiment, the fringes are displaced by a distance y when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate is replaced by another plate of the same thickness, the shift of fringes is $\frac{3}{2}y$. The refractive index of the second plate is
 (A) 1.75 (B) 1.50 (C) 1.25 (D) 1.00
- A point source of light is placed at depth h below the surface of a large and deep lake. The percentage of light energy that escapes directly from the water surface is ($\mu_w = 4/3$).
 (A) Zero (B) 10% (C) 17% (D) 25%
- If the stationary proton and α -particle are accelerated through same potential difference, the ratio of de-broglie's wavelengths will be :
 (A) 2 (B) 1 (C) $2\sqrt{2}$ (D) None of these
- Four charges are placed each at a distance ' a ' from origin. The dipole moment of configuration is



- (A) $2qa\hat{j}$ (B) $3qa\hat{j}$ (C) $2aq(\hat{i} + \hat{j})$ (D) $2aq(\hat{i} - \hat{j})$
- The equivalent resistance between the points A and B is

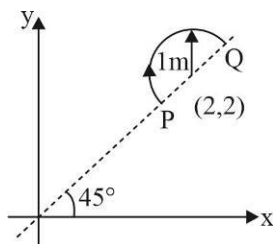


- (A) $\frac{36}{7}\Omega$ (B) 10Ω (C) $\frac{85}{7}\Omega$ (D) 18Ω
- In the circuit shown in the figure, charge stored in the capacitor of capacity $5\mu F$ is

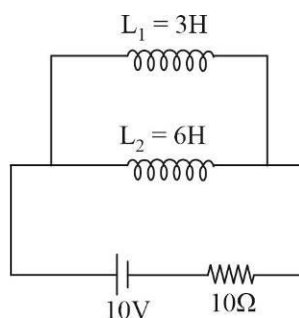


- (A) $60\mu C$ (B) $20\mu C$ (C) $30\mu C$ (D) Zero

8. A uniform magnetic field $B = (3\hat{i} + 4\hat{j} + \hat{k})$ exists in region of space. A semicircular wire of radius 1 m carrying current 1A having its centre at (2, 2, 0) is placed in x-y plane as shown in the figure. The force on semicircular wire will be :



- (A) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$ (B) $\sqrt{2}(\hat{i} - \hat{j} + \hat{k})$ (C) $\sqrt{2}(\hat{i} + \hat{j} - \hat{k})$ (D) $\sqrt{2}(-\hat{i} + \hat{j} + \hat{k})$
9. Two inductor coils of self inductance 3H and 6H respectively are connected with a resistance 10Ω and a battery 10 V as shown in the figure. The ratio of total energy stored at steady state in the inductors to that of heat developed in resistance in 10 seconds at the steady state is (neglect mutual inductance between L_1 & L_2)



- (A) $\frac{1}{10}$ (B) $\frac{1}{100}$ (C) $\frac{1}{1000}$ (D) 1
10. A cylindrical tank of height 50 cm is completely filled with water. On its vertical side two small holes are there, such that water flowing out of both the holes strikes the ground at same point. If first hole is at depth 30 cm from the top, then the depth of second hole from top will be :
- (A) 10 cm (B) 20 cm (C) 25 cm (D) 40 cm
11. A ball submerges completely and floats inside a liquid of density ρ . If same ball is dropped in a liquid of density 4ρ , then the percentage volume of the ball immersed inside this new liquid will be?
- (A) 25% (B) 50% (C) 75% (D) 20%
12. When 50 g of liquid P at temperature 80°C is mixed with 100 g of liquid Q at 40°C , the temperature of mixture becomes 60°C . If 25 g of liquid P at 100°C is added to 100 g of liquid Q at 25°C , then the temperature of mixture will be :
- (A) 70°C (B) 60°C (C) 40°C (D) 50°C
13. A metallic sphere cools from 60°C to 52°C in 10 minutes. If the surrounding temperature is 16°C , then the temperature of sphere after the next 10 minutes is close to :
- (A) 44°C (B) 45°C (C) 49°C (D) 51°C
14. Two light aluminium rods of same length are suspended from a ceiling. First rod has load W hanging from it, while second rod has load 2W hanging from it. If the ratio of energy stored per unit volume of first rod to second rod is 1:64, then the ratio of diameter of first rod to second rod will be :
- (A) 4:1 (B) 2:1 (C) 1:2 (D) 1:4

15. A heat engine released 1020 J of heat and performed 900 J of useful work. A cannot engine operating between the same temperature ranges as this engine is 30% more efficient. If temperature of source is 700 K, temperature of sink is
(A) 390 K **(B)** 280 K **(C)** 460 K **(D)** 170 K
16. A container has 0.6 moles of a monoatomic gas at 27°C. If 0.09 moles of the same gas at 227°C is added to it, the final equilibrium temperature of the container will be around.
(A) 50°C **(B)** 70°C **(C)** 30°C **(D)** 100°C
17. Time period of a pendulum in a stationary lift is 2 seconds. Now the lift is moving up at a speed of 30 m/s but slowing down at a rate of $5m/s^2$. What is the time period of the pendulum now?
(A) 1.6 s **(B)** 2.8 s **(C)** 1.1 s **(D)** 2.3 s
18. For a particle undergoing SHM, displacement from mean position is given according to the law : $x(t) = (10cm)\sin\left(\frac{2\pi t}{45}\right)$, where t is in seconds. The mean position of particle is located at $x = 2cm$.
 The ratio of KE and PE of particle when it is at $x = -3cm$ is (Assume PE at mean position is zero)
(A) 2 **(B)** 0.33 **(C)** 0.5 **(D)** 3
19. A transverse wave, $y = -0.5\sin(x + 45t)$ is produced in a wire of mass per unit length 0.12 g/cm, where x is in meters and t is in seconds. Then the expected value of tension in the wire is _____.
(A) 16.7 N **(B)** 24.3 N **(C)** 29.2 N **(D)** 26.7 N
20. In the experiment of resonance column with diameter 10cm, and length 70cm, first resonance is observed when tube has 55cm of water. Speed of sound at given temperature is 330 m/s. Find the frequency of tuning fork.
(A) 400 Hz **(B)** 390 Hz **(C)** 510 Hz **(D)** 460 Hz

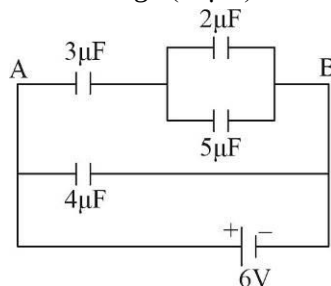
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SECTION-2

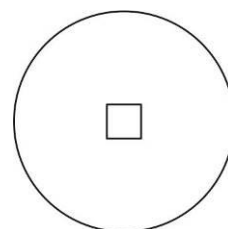
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21. For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Calculate the refractive index of prism.
22. The wavelength of K_{α} X-rays produced by an X-ray tube is $0.76 \text{ \AA}$. The atomic number of the anticathode material is _____.
23. Four point charges $q, -q, 2Q$ & Q are placed in order at the corners A, B, C and D of a square. If the electric field at the midpoint of CD is zero then the value of q/Q is $\frac{5\sqrt{5}}{x}$. Find the value of x .
24. A circuit is shown in the figure. Find the charge (in μC) on the condenser having a capacity of $5\mu\text{F}$.

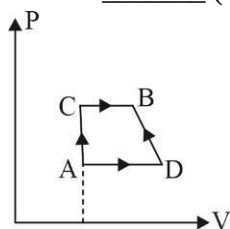


25. Consider the two loops with common center and on the same plane as shown in the figure. The large circular loop has radius $R = 10 \text{ cm}$ and the small square loop has side $a = 1 \text{ cm}$. There is a current circulating in the large loop given by $i(t) = i_0 t / \tau$ with $i_0 = \frac{2}{\pi} \times 100 \text{ A}$, $\tau = 2 \times 10^{-2} \text{ sec}$.



Emf (in microvolts) induced in the small loop is _____.

26. If a spherical blackbody's temperature is increased from T to $2T$, while its volume is reduced to $1/8$ times of its initial value, then the ratio of new radiant power to the initial radiant power is x . Then the value of x is _____.
27. A liquid drop has surface energy U . It breaks into n equal small droplets. If the new surface energy after division becomes $5U$, then the value of n is _____.
28. A monoatomic gas is taken from A to B through two different processes ACB and ADB. Gas gains 50 J of internal energy in process CB and 70 J of heat is given in process AC. If work done by gas in process ADB is 60 J , then heat flow in process ADB is _____ (in Joule).



29. For a dampened spring mass system having mass = 200 g and decay constant = 10 g/s, the time in which amplitude drops to $1/5^{\text{th}}$ of its initial value is close to 2^x . (take $\ln 5 = 1.61$), find x.
30. A galaxy is moving away from the earth at a speed of 410 kms^{-1} . The shift in the wavelength of a red line at 600 nm is $x \times 10^{-10} \text{ m}$. The value of x, to the nearest integer, is _____. [Take the value of speed of light c, as $3 \times 10^8 \text{ ms}^{-1}$]

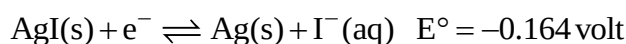
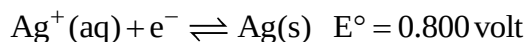
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PART - II : CHEMISTRY**120 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. What will be the void space in closest packing of 'n' spheres of radius '1' unit, 'n' spheres of radius 0.414 units and 2n spheres of radius 0.225 units?
 (A) 26% (B) 12% (C) 19% (D) 32%

2. Calculate the value of solubility product for solid AgI by using following data :

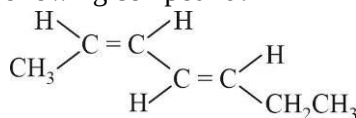


- (A) 5×10^{-17} (B) 6×10^{-15} (C) 2×10^{-14} (D) 4×10^{-12}
3. Two radioactive nuclides A and B have half lives in the ratio 2 : 3 respectively. An experiment is made with one mol of each of them. Calculate the molar ratio of A and B after a time interval of three times of half life of A
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

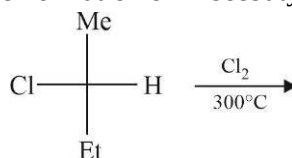
4. Which of the following is lyophilic?

- (A) Gelatin sol (B) Silver sol (C) Sulphur sol (D) As_2S_3 sol

5. What is the IUPAC name of the following compound?



- (A) hepta-(2E, 4Z)-diene (B) hepta-(2E, 4E)-diene
 (C) hepta-(2Z, 4E)-diene (D) hepta-(2Z, 4Z)-diene
6. Pick the correct statement for monochlorination of R-secbutyl chloride.

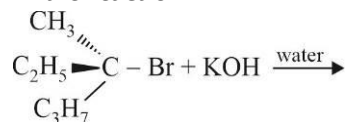


- (A) There are four possible products ; three are optically active and one is optically inactive
 (B) There are five possible products ; three are optically inactive & two are optically active
 (C) There are five possible products ; two are optically inactive & three are optically active
 (D) There are four possible products ; two are optically inactive & two are optically active

7. (Di-nitro product)
 Product (C) of this reaction is

- (A) (B) (C) (D)

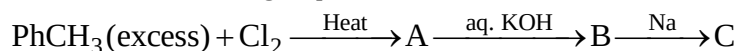
8. In the reaction



If the concentration of both the reactants is doubled, the rate of the reaction will

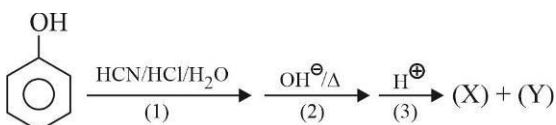
- (A) Double (B) Quadruple
(C) Be reduced to one-fourth (D) Remain unchanged
9. Ethylene glycol on treatment with PI_3 mainly gives :
- (A) Ethylene (B) Ethylene iodide (C) Ethyl iodide (D) Ethane

10. Consider the following sequence of reaction



$\text{A} + \text{C} \xrightarrow{\text{Heat}} \text{D}$. Product D is :

- (A) PhCH_2OPh (B) $\text{PhCH}_2\text{OCH}_2\text{Ph}$
(C) $\text{PhCH}_2\text{CH}_2\text{Ph}$ (D) $\text{Ph}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{Ph}$

- 11.
- 
- $$\text{C}_6\text{H}_5\text{OH} \xrightarrow{(1) \text{ HCN/HCl/H}_2\text{O}} \text{Intermediate} \xrightarrow{(2) \text{ OH}^-/\Delta} \text{Intermediate} \xrightarrow{(3) \text{ H}^+} (\text{X}) + (\text{Y})$$

X gives white turbidity with Lucas reagent instantly.

X and Y both turn blue litmus solution red. Y can be :

- (A) p-Hydroxy benzoic acid (B) p-Hydroxy benzaldehyde
(C) m-Hydroxy benzoic acid (D) p-Hydroxy benzyl alcohol
12. A compound $\text{A}(\text{C}_5\text{H}_{10}\text{Cl}_2)$ on hydrolysis gives $\text{C}_5\text{H}_{10}\text{O}$ which reacts with NH_2OH , forms iodoform but does not give Fehling test. A is

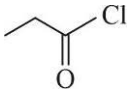
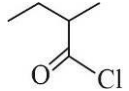
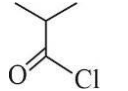
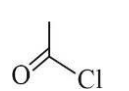
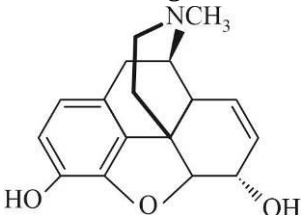
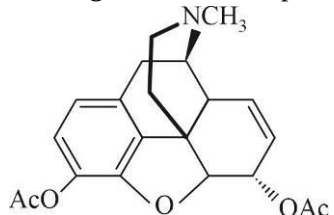
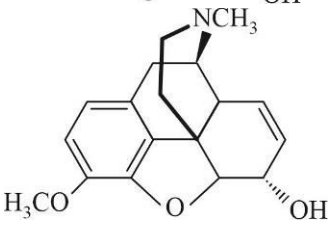
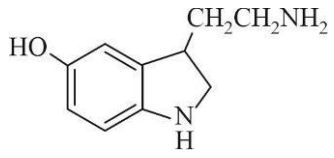
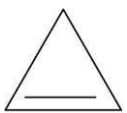
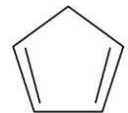
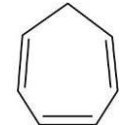
- (A) $\text{CH}_3-\overset{\text{Cl}}{\underset{\text{Cl}}{\text{C}}}-\text{CH}_2-\text{CH}_2-\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2-\overset{\text{Cl}}{\underset{\text{Cl}}{\text{C}}}-\text{CH}_2\text{CH}_3$
(C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\overset{\text{Cl}}{\text{CH}}-\text{Cl}$ (D) $\text{CH}_3-\overset{\text{Cl}}{\text{CH}}-\overset{\text{Cl}}{\text{CH}}-\text{CH}_2\text{CH}_3$

13. The compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess of
- CO_2
- is bubbled through aqueous solutions of Y and Z is formed. Z on gentle heating gives back X. The compound X is:

- (A) CaCO_3 (B) Na_2CO_3 (C) $\text{Ca}(\text{NO}_3)_2$ (D) K_2CO_3

14. When acidified
- KMnO_4
- is added to hot oxalic acid solution, the decolorization is slow in the beginning but becomes very rapid after some time. This is because:

- (A) Mn^{+2} acts as auto catalyst (B) CO_2 is formed as the product
(C) Reaction is exothermic (D) MnO_4^- catalyzes the reaction

15. Which of the following order is not correct?
- (A) $F^- < Cl^- < Br^- < I^-$ (reducing nature)
 (B) $F^- > Cl^- > Br^- > I^-$ (hydration energy)
 (C) $Cl_2 > F_2 > Br_2 > I_2$ (bond dissociation energy)
 (D) $F_2 > Cl_2 > Br_2 > I_2$ (reactivity)
16. The correct order of magnetic moment (spin only values in B.M.) is:
 I. $[MnBr_4]^{2-}$ II. $[Fe(CN)_6]^{4-}$ III. $[CoBr_4]^{2-}$
 (A) $II > III > I$ (B) $I > II > III$ (C) $II > I > III$ (D) $I > III > II$
17. (A) $\xrightarrow{NH_3}$ (B) $\xrightarrow{KOH/Br_2}$ $CH_3 - \overset{\overset{CH_3}{|}}{CH} - NH_2$;
 structure of (A) is
 (A)  (B)  (C)  (D) 
18. Which of the following is a component of photochemical smog?
 (A) C_2H_4 (B) N_2O
 (C) C_3H_4O (Cyclic ketone) (D) C_3H_4O (Unsaturated aldehyde)
19. Which of the following is the correct structure of a narcotic analgesic called morphine?
- (A)  (B) 
- (C)  (D) 
20. Which of the following will react readily with aqueous solution of Na_2CO_3 ?
- (A)  (B)  (C)  (D) 

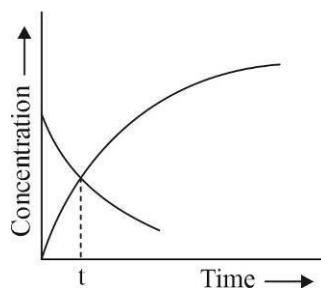
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SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 questions have to be attempted.

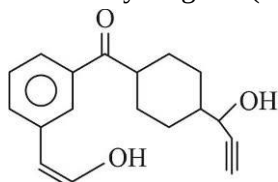
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21. Air is saturated with water vapors having 1230 torr pressure. If volume is doubled, the new pressure in container at equilibrium becomes x torr. Find the value of $\frac{x}{126}$ (V.P. of $H_2O = 30$ torr).
22. A 500 ml of aqueous NaCl solution is electrolyzed for 1 minute using a current of 8mA at $25^\circ C$. What is the pH of the final solution? (Use $1F = 96000$ C)
23. Consider the following first order decomposition process $A_n \rightarrow nA$



Here, t corresponds to the time at which $\left(\frac{1}{6}\right)^{th}$ of reactant is decomposed. The value of 'n' is _____.

24. An alkyne has molar mass 96. How many different isomers (excluding stereochemical designation) are possible considering all of them to be internal alkynes?
25. How many reagents (i to vii) can successively react with the following compound?



- | | | |
|------------------------|----------------------|--|
| (i) $NaHCO_3$ | (ii) 2, 4- DNP | |
| (iii) Na metal | (iv) $AgNO_3 + OH^-$ | |
| (v) Fehling's solution | (vi) Lucas reagent | |
| (vii) Br_2 / H_2O | | |
26. Number of oxide ores out of the following are:
Cuprite, Galena, Cassiterite, Magnetite, Bauxite, Carnallite, Calamine
27. How many of the following vitamin(s) are water soluble?
- | | | |
|------------------------|-------------------|--------------------|
| (i) Vitamin A | (ii) Vitamin B | (iii) Vitamin C |
| (iv) Vitamin D | (v) Vitamin B_6 | (vi) Vitamin E |
| (vii) Vitamin B_{12} | (viii) Vitamin K | (ix) Vitamin B_2 |

28. How many of the following amines will undergo diazotisation?
tert-butylamine, Ethanamine, Aniline, N-methylaniline,
(I) (II) (III) (IV)
p-toluidine, m-chloroaniline, 2-phenylethanamine
(V) (VI) (VII)
29. From the following list, the number of acidic oxides is 'a', the number of neutral oxides is 'b' and the number of amphoteric oxides is 'c'.
 CO_2 , CO, GeO_2 , GeO, SiO_2 , SnO_2 , SnO, PbO_2 , PbO
Find the value of $(a + b - c)$.
30. Number of hydrogen atoms in a repeating unit of a polymer used to make unbreakable crockery is _____. (Integer answer)

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**120 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Axis of a parabola lies along x -axis. If its vertex and focus are at distances 2 and 4 respectively from the origin, on the positive x -axis then which of the following points does not lie on it?
 (A) $(6, 4\sqrt{2})$ (B) $(4, -4)$ (C) $(5, 2\sqrt{6})$ (D) $(8, 6)$
- If $f(x) = \min(2 \sin x, 1 - \cos x, 1)$, then $\int_0^{\pi} f(x) dx =$
 (A) $\sqrt{3} - 1 + \frac{5\pi}{6}$ (B) $\sqrt{3} - 1 + \frac{2\pi}{3}$ (C) $1 - \sqrt{3} + \frac{2\pi}{3}$ (D) $1 - \sqrt{3} + \frac{5\pi}{6}$
- A 2×2 matrix is formed with entries from the set $\{0, 1\}$. The probability that it is singular is :
 (A) $\frac{5}{8}$ (B) $\frac{1}{8}$ (C) $\frac{5}{16}$ (D) None of these
- Consider the equation $4^{\sin x} + 3^{\sec y} = 11$ and $5(16)^{\sin x} - 2(3)^{\sec y} = 2$. Then, the exhaustive values of x and y respectively are :
 (A) $x = n\pi + \frac{\pi}{6}, y = 2m\pi + \frac{\pi}{3}; m, n \in \mathbb{Z}$
 (B) $x = n\pi - \frac{\pi}{6}, y = 2m\pi + \frac{\pi}{3}; m, n \in \mathbb{Z}$
 (C) $x = n\pi + (-1)^n \frac{\pi}{6}, y = 2m\pi \pm \frac{\pi}{3}; m, n \in \mathbb{Z}$
 (D) $x = n\pi + (-1)^n \frac{\pi}{6}, y = 2m\pi - \frac{\pi}{3}; m, n \in \mathbb{Z}$
- Let $x = (5\sqrt{2} + 7)^{19}$, then $x\{x\}$ ($\{x\}$ denotes the fractional part of x) is equal to :
 (A) 2^{19} (B) 3^{19} (C) 0 (D) 1
- The locus of the middle points of the normal chords of the parabola, $y^2 = 4ax$ is :
 (A) $x + 2a = \frac{y^2}{2a} + \frac{4a^3}{y^2}$ (B) $x + 2a = \frac{y^2}{2a} - \frac{4a^3}{y^2}$
 (C) $x - 2a = \frac{y^2}{2a} + \frac{4a^3}{y^2}$ (D) None of these
- If $\int \frac{2 \cos x - \sin x + \lambda}{\cos x + \sin x - 2} dx = A \ln |\cos x + \sin x - 2| + Bx + C$. Then, ordered triplet (A, B, λ) is :
 (A) $\left(\frac{1}{2}, \frac{3}{2}, -1\right)$ (B) $\left(\frac{3}{2}, \frac{1}{2}, -1\right)$ (C) $\left(\frac{1}{2}, -1, -\frac{3}{2}\right)$ (D) None of these

8. In an isosceles triangle ABC, the coordinates of the point B and C on the base BC are respectively (2, 1) and (1, 2). If the equation of the line AB is $y = \frac{x}{2}$, then equation of the line AC is :
- (A) $2y = x + 3$ (B) $y = 2x$ (C) $y = \frac{1}{2}(x - 1)$ (D) $y = x - 1$
9. If $y^2 = p(x)$, then $2 \frac{d}{dx} \left(y^3 \frac{d^2 y}{dx^2} \right)$.
- (A) $p'''(x) + p(x)$ (B) $p''(x) \cdot p'''(x)$ (C) $p(x) \cdot p'''(x)$ (D) None of these
10. The area of the region bounded by the curve $y = \frac{16 - x^2}{4}$ and $y = \sec^{-1}[-\sin^2 x]$ (where $[.]$ denotes greatest integer function) is :
- (A) $\frac{1}{3}(4 - \pi)^{3/2}$ (B) $8(4 - \pi)^{3/2}$ (C) $\frac{8}{3}(4 - \pi)^{3/2}$ (D) $\frac{8}{3}(4 - \pi)^{1/2}$
11. The value of $\lim_{n \rightarrow \infty} \left(\sin^4 x + \frac{1}{4} \sin^4 2x + \dots + \frac{1}{4^n} \sin^4 (2^n x) \right)$ is :
- (A) $\sin^2 x$ (B) $\sin^4 x$ (C) $\cos^2 x$ (D) None of these
12. Let $a_r, r = 1, 2, 3, 4$ represent four positive real numbers other than unity such that $\left(\log_e a_r, \frac{1}{\log_e a_r} \right)$ is a point on a circle whose abscissa of ends of diameter are α and $-\alpha$ ($\alpha \neq 0$), then the minimum value of $a_1 + a_2 + a_3 + a_4$ is :
- (A) 2 (B) 3 (C) 4 (D) 5
13. A plane $2x + 3y + 6z = 12$ intersects the x-axis, y-axis and z-axis at A, B, C respectively. The orthocentre of the triangle ABC is :
- (A) $\left(\frac{24}{49}, \frac{36}{49}, \frac{72}{49} \right)$ (B) $\left(\frac{36}{49}, \frac{72}{49}, \frac{27}{49} \right)$ (C) $\left(\frac{24}{49}, \frac{16}{49}, \frac{1}{49} \right)$ (D) $\left(\frac{36}{49}, \frac{72}{49}, \frac{24}{49} \right)$
14. Let $\frac{\tan\left(\frac{\pi}{4} + \alpha\right)}{5} = \frac{\tan\left(\frac{\pi}{4} + \beta\right)}{3} = \frac{\tan\left(\frac{\pi}{4} + \gamma\right)}{2}$. Then $12 \sin^2(\alpha - \beta) + 15 \sin^2(\beta - \gamma) - 7 \sin^2(\gamma - \alpha)$ is equal to :
- (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 1 (D) 0
15. The general solution of the differential equation $y \left(xy - (x^2 - y^2)^2 \right) dx = \left\{ y^3 - x(x^2 - y^2)^2 \right\} dy$ is :
[c is arbitrary constant]
- (A) $\frac{y}{x} + \frac{1}{x^2 - y^2} = c$ (B) $\frac{y}{x} - \frac{1}{2} \ln \left| \frac{x - y}{x + y} \right| = c$
- (C) $\frac{2x}{y} + \frac{1}{x^2 - y^2} = c$ (D) $\frac{2x}{y} - \frac{1}{x^2 - y^2} = c$

16. The perpendicular distance of point $P(0, -1, 3)$ from the straight line passing through $A(1, -3, 2)$ and $B(2, -1, 4)$ is :
- (A) $\frac{\sqrt{29}}{3}$ units (B) $\frac{\sqrt{23}}{3}$ units (C) $\frac{\sqrt{19}}{3}$ units (D) $\frac{\sqrt{17}}{3}$ units
17. The function $f(x) = \sin^4 x + \cos^4 x$ increases, if :
- (A) $0 < x < \frac{\pi}{8}$ (B) $\frac{\pi}{4} < x < \frac{3\pi}{8}$ (C) $\frac{3\pi}{8} < x < \frac{5\pi}{8}$ (D) $\frac{5\pi}{8} < x < \frac{3\pi}{4}$
18. If range of $f(x) = \cos x$, $x \in \left(-\frac{\pi}{3}, \frac{\pi}{6}\right]$ is $(a, b]$ then :
- (A) $a + b = \frac{3}{2}$ (B) $b - a = 1 - \frac{\sqrt{3}}{2}$ (C) $a^2 + b^2 = \frac{5}{6}$ (D) $a^2 + b^2 = \frac{7}{4}$
19. The value of $\int_1^e \left(\frac{1}{x} - x + x \ln x \right) \sin x \, dx$ is :
- (A) $\sin e - \cos 1$ (B) $\cos e - \sin 1$ (C) $\sin e + \cos 1$ (D) $\cos e + \sin 1$
20. Consider the differential equation $y^2 dx + \left(x - \frac{1}{y} \right) dy = 0$. If $y(1) = 1$, then x is given by:
- (A) $4 - \frac{2}{y} - \frac{e^{1/y}}{e}$ (B) $3 - \frac{1}{y} + \frac{e^{1/y}}{e}$ (C) $1 + \frac{1}{y} - \frac{e^{1/y}}{e}$ (D) None of these

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SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 questions have to be attempted.

The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25, 0.08)

21. If A, B, C and D are four points in space, and $\left| \overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD} \right| = k$. (Area of $\triangle ABC$) then k is equal to _____.
22. The value of integral $\int_{-\pi/4}^{3\pi/4} \frac{\sin x + \cos x}{e^{x-\frac{\pi}{4}} + 1} dx$ is equal to $\sqrt{2k}$, then find the value of k .
23. The value of $\lim_{x \rightarrow \infty} \left(\sqrt{3x^2 + \sqrt{3x^2 + \sqrt{3x^2}}} - \sqrt{3x^2} \right)$ is _____.
24. Let f be a function defined on the interval $[0, 2\pi]$ such that $\int_0^x (f'(t) - \sin 2t) dt = \int_x^0 f(t) \tan t dt$ and $f(0) = 1$. Then the maximum value of $f(x)$ is _____.
25. Let $y = f(x) = 4x^3 + 2x - 6$, then the value of $\int_0^2 f(x) dx + \int_0^{30} f^{-1}(y) dy$ is equal to _____.
26. Let α_r where $r = 1, 2, 3, 4, \dots, 100$ be the roots of $\sum_{k=0}^{100} (z)^k = 0$. If $\sum_{r=1}^{100} \frac{1}{(\alpha_r - 1)} = 10\lambda$, then $|\lambda|$ is equal to _____.
27. A line is drawn at an angle θ with positive direction of x -axis. It intersects the parabola $y^2 = 8x$ at A and B. If AB is a normal chord which subtends a right angle at the vertex of the parabola, then $\tan^2 \theta$ equals _____.
28. Let f be a continuous and differentiable function in (a, b) , if $f(x), f'(x) \geq x\sqrt{1 - (f(x))^4}$ and $\lim_{x \rightarrow a^+} (f(x))^2 = 1$ and $\lim_{x \rightarrow b^-} (f(x))^2 = \frac{1}{2}$. Then the minimum value of $[a^2 - b^2] + 80$ is _____. ([.] denotes represent greatest integer function)
29. Let O be an interior point $\triangle ABC$ such that $\overrightarrow{OA} + 2\overrightarrow{OB} + 3\overrightarrow{OC} = \vec{0}$ then ratio of the area of $\triangle ABC$ to the area $\triangle AOC$ is _____.

30. A purse contains 2 dice. One is a normal fair die, while the other has 2 ones, 2 threes, and 2 fives. A die is picked up and rolled. Because of some secret magnetic attraction of the unfair die, there is 75% chance of picking the unfair die and a 25% chance of picking a fair die. The die is rolled and shows up the face 3. The probability that a fair die was picked up, is a/b where a and b are coprime then $a + b$ equals to _____.

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